

Workbook: Muscle Fascicle Arrangements

Week 4 workbook • BIO 004 Human Anatomy • Dr. Sharilyn Rennie • Solano Community College

A focused workbook on the arrangement of fascicles in skeletal muscles, the architectural patterns, and how each pattern relates to function. Pair with the broader muscle microanatomy workbook.

How to use this workbook: Read each question carefully and write your answer in the space provided. Work without looking up the answer first. Once you have written your answer, check it against the online workbook by clicking *Reveal* on that question. Star or circle anything you missed and come back to it later in the week.

Fascicle patterns • 6 questions

Q01 • DOK 1 • Recall

What is a fascicle, and what wraps it?

Q02 • DOK 1 • Recall

Describe parallel fascicle arrangement and give an example.

Q03 • DOK 1 • Recall

Describe convergent fascicle arrangement and give an example.

Q04 · DOK 2 · Compare

Compare unipennate, bipennate, and multipennate muscle architecture.

Q05 · DOK 1 · Recall

Describe circular fascicle arrangement and give two examples.

Q06 · DOK 3 · Reason

Why does pennate architecture produce more force per unit volume than parallel architecture, and what does the body trade for that gain?